Name: ____ Date: ___________________

Point values are assigned for each question. Points earned: ____ / 100, = ____ %

1. Find a tight upper bound for $f(n) = n^5 + 8$. Write your answer here: _______ (4 points)

   Prove your answer by giving values for the constants c and n_0. Choose the smallest integer value possible for c. (4 points)

2. Find an asymptotically tight bound for $f(n) = n^3 + 5n$. Write your answer here: _______ (4 points)

   Prove your answer by giving values for the constants c_1, c_2, and n_0. Choose the tightest integer values possible for c_1 and c_2. (6 points)

3. Is $2n^2 - n \in \Omega(n^3)$? Circle your answer: yes / no. (2 points)

   If yes, prove your answer by giving values for the constants c and n_0. Choose the smallest integer value possible for c. If no, derive a contradiction. (4 points)

4. Write the following asymptotic efficiency classes in **increasing** order of magnitude.
   $O(\lg n^2)$, $O(0.01^n)$, $O(20!\, n)$, $O(n \lg n)$, $O(n)$, $O(n!)$, $O(n^8)$, $O(\lg \lg n)$, $O(n^n)$, $O(n^{20} \lg n)$ (2 points each)

   ________, _________, _________, _________, ________, ________, ________, ________, _________, ________

5. Give the complexity of the following methods. Choose the most appropriate notation from among O, Θ, and Ω. (8 points each)

```
int function1(int n) {
    int count = 0;
    for (int i = 0; i <= n/2; i++) {
        for (int j = 1; j <= n; j *= 2) {
            count--;
        }
    }
    return count;
}
```

Answer: _________

```
int function2(int n) {
    int count = 0;
    for (int i = 1; i * i * i * i <= n; i++) {
        count++;
    }
    return count;
}
```

Answer: _________

```
int function3(int n) {
    int count = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            for (int k = 1; k <= n; k++) {
                for (int z = 1; z <= n; z++) {
                      count++;
                }
            }
        }
    }
    return count;
}
```

Answer: _________

```
int function4(int n) {
    int count = 0;
    for (int i = 1; i <= 2^n; i++) {
        for (int j = 1; j <= n * n; j++) {
            count++;
        }
        break;
    }
    return count;
}
```

Answer: _________

```
int function5(int n) {
    int count = 0;
    for (int i = 1; i <= n && i < 1; i++) {
        count++;
    }
    for (int j = 1; j <= n; j++) {
        count++;
    }
    return count;
}
```

Answer: _________